

A Conformally Invariant Gap Theorem in Yang–Mills Theory

Based on Gursky, Kelleher, Streets (2018)

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1 Introduction and Motivation

In this lecture, we will explore a significant advance in the theory of Yang–Mills connections over four-dimensional manifolds, focusing on a conformally invariant energy gap theorem. Yang–Mills theory in four dimensions plays a central role in both mathematics and physics. Critical points of the Yang–Mills energy functional,

$$E(\nabla) = \int_X |F_\nabla|^2 dV_g,$$

are called Yang–Mills connections. Among these, instantons—solutions with self-dual or anti-self-dual curvature—are energy minimizers and foundational objects in Donaldson theory and four-manifold topology.

Despite their centrality, not all Yang–Mills connections are instantons. A fundamental and long-standing question is whether one can understand the energy spectrum of Yang–Mills connections beyond the self-dual sector. In particular, if a connection is not an instanton, can we say something quantitative about how much higher its energy must be?

Previous works have established “gap theorems” under curvature positivity assumptions or ε -regularity frameworks. However, these conditions are restrictive. The main contribution

of Gursky, Kelleher, and Streets is to lift these constraints and provide a sharp, conformally invariant energy lower bound for non-self-dual Yang–Mills connections when the underlying manifold has positive Yamabe invariant.

2 The Main Theorem

The central result of the paper is a dichotomy: for any Yang–Mills connection on a closed, oriented four-manifold, either the connection is anti-self-dual, or it must obey a sharp lower bound involving the Yamabe invariant and the L^2 norms of the curvature and Weyl tensor.

Theorem 2.1 (The Main Theorem). *Let (X^4, g) be a closed, oriented four-manifold. Suppose ∇ is a Yang–Mills connection on a vector bundle E over X^4 with structure group $G \subset SO(E)$, and curvature F_∇ . Then one of the following must hold:*

- $F_\nabla^+ \equiv 0$; or
- F_∇^+ satisfies

$$Y([g]) \leq 3\gamma_1 \|F_\nabla^+\|_{L^2} + 2\sqrt{6} \|W^+\|_{L^2}, \quad (1)$$

where $\gamma_1 = \gamma_1(E) \leq \frac{4}{\sqrt{6}}$ is a constant which depends on the structure group of the bundle, and W^+ is the self-dual Weyl tensor.

Moreover, if equality holds in (1.1) then $[g]$ admits a Yamabe metric $\bar{g}$ with respect to which W^+ has constant norm, $\nabla_{\bar{g}} F_{\nabla_{\bar{g}}}^+ \equiv 0$, and

$$R_{\bar{g}} - 2\sqrt{6} |W^+|_{\bar{g}} = 3\gamma_1 |F_{\nabla}^+|_{\bar{g}}, \quad (2)$$

where $R_{\bar{g}}$ is the scalar curvature of $\bar{g}$. Furthermore, if $\gamma_1 > 0$ we must have $b_2^+(M^4) = 0$, whereas if $\gamma_1 = 0$ then all harmonic self-dual forms are parallel.

2.1 Overview of the Proof

The central innovation of the proof lies in reducing a Yang–Mills energy gap problem to a conformally invariant variational problem of Yamabe type.

The argument begins with a Bochner formula applied to the self-dual part F_∇^+ of the curvature of a Yang–Mills connection. Because F^+ is harmonic, it satisfies a Weitzenböck-type identity that includes terms involving $|\nabla F^+|^2$, the Weyl curvature W^+ , and a nonlinear Lie bracket $[F^+, F^+]$. These latter two terms are estimated using sharp algebraic inequalities, introducing a key constant γ_1 that depends only on the structure group.

Next, an improved Kato inequality for F^+ is applied, allowing the authors to transform the Bochner identity into a differential inequality for $|F^+|^{1/2}$. This inequality takes the schematic form:

$$L(|F^+|^{1/2}) \leq 0,$$

where L is a modified Yamabe-type operator. Here lies the key insight: this operator behaves like the classical conformal Laplacian and retains the same conformal covariance. Thus, the authors reinterpret the problem of bounding the Yang–Mills energy as a spectral problem for L , analogous to the Yamabe problem of finding constant scalar curvature metrics in a conformal class.

By testing L against suitable test functions (in particular, $|F^+|^{1/2}$ and the first eigenfunction of L), one obtains an integral inequality. Combining this with the variational characterization of the Yamabe invariant yields the estimate in the main theorem.

Thus, a sharp energy gap for non-self-dual Yang–Mills connections is derived by embedding the curvature term into a conformally covariant scalar curvature problem. This elegant translation from gauge theory to conformal geometry is the conceptual heart of the proof.

2.2 Preparatory Ingredients

Bochner-Weitzenböck Formula for Self-Dual Forms

Let (X^4, g) be a Riemannian manifold, and suppose $E \rightarrow X$ is a smooth vector bundle with connection ∇ . If $\omega \in \Lambda_+^2(\mathfrak{g}_E)$ is a harmonic two-form, one has

$$\frac{1}{2}\Delta|\omega|^2 = |\nabla\omega|^2 - \langle\omega, [F^+, \omega]\rangle - 2\langle\omega, W^+ \star \omega\rangle + \frac{1}{3}R|\omega|^2, \quad (3)$$

where with respect to local bases one has, for $P, Q \in \Lambda_+^2(\mathfrak{g}_E)$, the tensor $[P, Q] \in \Lambda_+^2(\mathfrak{g}_E)$ defined via

$$[P, Q]_{ij\alpha}^\beta := g^{kl} \left(P_{ik\delta}^\beta Q_{jl\alpha}^\delta - P_{ik\alpha}^\delta Q_{jl\delta}^\beta - P_{jk\delta}^\beta Q_{il\alpha}^\delta + P_{jk\alpha}^\delta Q_{il\delta}^\beta \right),$$

and

$$(W^+ \star \omega)_{ij\alpha}^\beta := g^{kp} g^{lq} W_{ijkl}^+ \omega_{pq\alpha}^\beta.$$

For the self-dual part of the curvature 2-form of a Yang–Mills connection F_∇^+ , since it is harmonic (i.e., both $dF^+ = 0$ and $d^*F^+ = 0$), the Bochner-Weitzenböck formula for bundle-valued self-dual 2-forms applies:

$$\frac{1}{2}\Delta|F^+|^2 = |\nabla F^+|^2 - \langle F^+, [F^+, F^+] \rangle - 2\langle F^+, W^+ \star F^+ \rangle + \frac{1}{3}R|F^+|^2. \quad (4)$$

This forms the analytic starting point of the proof. Each term will be carefully bounded or interpreted geometrically.

Estimate for the Weyl Tensor Term

Let (M^4, g) be a Riemannian manifold, and suppose $E \rightarrow M$ is a smooth vector bundle with connection ∇ . If $\omega \in \Lambda_+^2(\mathfrak{g}_E)$ is a two-form, one has

$$|\langle\omega, W^+ \star \omega\rangle| \leq \frac{2}{\sqrt{6}}|W^+||\omega|^2. \quad (5)$$

Thus the action of the self-dual Weyl curvature W^+ on F^+ is

$$|\langle F^+, W^+ \star F^+ \rangle| \leq \frac{2}{\sqrt{6}} |W^+| |F^+|^2.$$

This allows one to extract a bound involving the L^2 -norm of the Weyl tensor in the eventual inequality.

Lie-Algebraic Estimate

The nonlinearity $\langle F^+, [F^+, F^+] \rangle$ arises from the curvature bracket. To control this term, one defines two universal constants depending on the Lie algebra $\mathfrak{g} \subset \mathfrak{so}_E$:

$$\begin{aligned} \gamma_0 &:= \sup_{A, B \in \mathfrak{g} \setminus \{0\}} \frac{|[A, B]|}{|A||B|}, \\ \gamma_1 &:= \sup_{\omega \in \Lambda_+^2(\mathfrak{g}) \setminus \{0\}} \frac{\langle \omega, [\omega, \omega] \rangle}{|\omega|^3}. \end{aligned}$$

On the one side, $\gamma_0 \leq \sqrt{2}$, with the supremum defining γ_0 attained by pairs A, B which are simultaneously equivalent to a pair of Pauli matrices, i.e.

$$A = \left(\begin{array}{cc|cc} 0 & t & 0 & 0 \\ -t & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \\ 0 & -t & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array} \right), \quad B = \left(\begin{array}{cc|cc} 0 & -s & 0 & 0 \\ 0 & 0 & s & 0 \\ \hline s & 0 & 0 & 0 \\ 0 & -s & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array} \right).$$

In the case $\mathfrak{g} = \mathfrak{so}_3$ one has $\gamma_0 = 1$.

On the other side, for $P \in \Lambda_+^2(\mathfrak{g}_E)$, one has

$$\|[P, P]\| \leq \frac{2}{\sqrt{3}} \gamma_0 |P|^2. \quad (6)$$

Improved Kato Inequality

For harmonic 2-forms valued in the Lie algebra bundle, one has the sharp Kato-type inequality:

$$|\nabla F^+|^2 \geq \frac{3}{2} |d|F^+|^2. \quad (7)$$

The Classical and Modified Yamabe Problem

The classical Yamabe problem seeks to find a conformal metric $g' = u^2 g$ on X^4 such that $R_{g'} = \text{const.}$ The key tool in this setting is the conformal Laplacian (or Yamabe operator):

$$L_g := -6\Delta + R,$$

which satisfies the conformal covariance:

$$L_{g'}(f) = u^{-3}L_g(uf), \quad \text{for } g' = u^2g.$$

The sign of the first eigenvalue of L_g :

$$\lambda_1(L_g) = \inf_{\phi \in C^\infty, \phi \neq 0} \frac{\int_X \phi L_g \phi dV_g}{\int_X \phi^2 dV_g}.$$

is a conformal invariant and determines the sign of scalar curvature realizable within $[g]$.

2.3 Modified Yamabe Problem and Conformal Covariance

By applying the Bochner–Weitzenböck formula (4), along with the estimate for the Weyl tensor term and the definition of γ_1 , we arrive at

$$\frac{1}{2}\Delta|F^+|^2 \geq |\nabla F^+|^2 + \frac{1}{3}(R - 2\sqrt{6}|W^+| - 3\gamma_1|F^+|)|F^+|^2. \quad (8)$$

By the Leibniz rule, away from the zero locus of F^+ we have

$$\frac{1}{2}\Delta|F^+|^2 = |F^+|\Delta|F^+| + |\nabla|F^+||^2,$$

hence by Improved Kato inequality (7),

$$\Delta|F^+| \geq \frac{1}{2} \frac{|\nabla|F^+||^2}{|F^+|} + \frac{1}{3}(R - 2\sqrt{6}|W^+| - 3\gamma_1|F^+|)|F^+|.$$

It follows that

$$\Delta|F^+|^{1/2} \geq \frac{1}{6}(R - 2\sqrt{6}|W^+| - 3\gamma_1|F^+|)|F^+|^{1/2} \quad (9)$$

off the zero locus of F^+ , or in the sense of distributions.

We next exploit (9) in conjunction with a modified Yamabe problem. In particular, given a metric $\hat{g} \in [g]$ we define the **modified scalar curvature**:

$$\Phi_{\hat{g}} = R_{\hat{g}} - 2\sqrt{6}|W^+|_{\hat{g}} - 3\gamma_1|F^+|_{\hat{g}}.$$

If we drop the Weyl and F^+ -terms then Φ is just the scalar curvature. Moreover, because these terms transform by scalings of the same weight under conformal changes of metric, the transformation law for Φ is essentially the same as the scalar curvature. More precisely, if we define the natural generalization of the conformal Laplacian, i.e. **modified Yamabe operator** by

$$L = -6\Delta + \Phi,$$

then given $\hat{g} = u^2g$ it follows that

$$\Phi_{\hat{g}} = u^{-3}L_g u. \quad (10)$$

The verification of this formula is as follows.

In dimension 4, scalar curvature transforms under a conformal change $\hat{g} = u^2 g$ as

$$R_{\hat{g}} = u^{-2} (R_g - 6u^{-1} \Delta_g u) = u^{-3} (-6\Delta_g u + R_g u).$$

Under conformal change, $|W^+|$ rescales as a 2-form:

$$|W^+|_{\hat{g}} = u^{-2} |W^+|_g.$$

Similarly, since F^+ is a self-dual 2-form with values in $\mathfrak{g}_E$, its norm transforms as:

$$|F^+|_{\hat{g}} = u^{-2} |F^+|_g.$$

Putting together the above transformations:

$$\begin{aligned} \Phi_{\hat{g}} &= R_{\hat{g}} - 2\sqrt{6}|W^+|_{\hat{g}} - 3\gamma_1|F^+|_{\hat{g}} \\ &= u^{-3} (-6\Delta_g u + R_g u) - 2\sqrt{6}u^{-2}|W^+|_g - 3\gamma_1 u^{-2}|F^+|_g \\ &= u^{-3} \left(-6\Delta_g u + R_g u - 2\sqrt{6}|W^+|_g u - 3\gamma_1|F^+|_g u \right) \\ &= u^{-3} (-6\Delta_g u + \Phi_g u) = u^{-3} L_g u. \end{aligned}$$

Moreover, L is conformally covariant:

$$L_{\hat{g}}\phi = u^{-3} L_g(u\phi). \quad (11)$$

The verification of this formula is as follows.

From the known transformation of the Laplacian on scalar functions in dimension 4, we have

$$-6\Delta_{\hat{g}}\phi = -6u^{-2} (\Delta_g \phi + 2u^{-1} \langle du, d\phi \rangle_g).$$

Next compute $L_g(u\phi)$,

$$\begin{aligned} L_g(u\phi) &= -6\Delta_g(u\phi) + \Phi_g u\phi \\ &= -6(\phi\Delta_g u + u\Delta_g \phi + 2\langle du, d\phi \rangle) + \Phi_g u\phi. \end{aligned}$$

Then

$$\begin{aligned} u^{-3} L_g(u\phi) &= u^{-3} [-6\phi\Delta_g u - 6u\Delta_g \phi - 12\langle du, d\phi \rangle + \Phi_g u\phi] \\ &= -6u^{-2} (\Delta_g \phi + 2u^{-1} \langle du, d\phi \rangle) + \phi u^{-3} (-6\Delta_g u + \Phi_g u) \\ &= -6\Delta_{\hat{g}}\phi + \phi\Phi_{\hat{g}} = L_{\hat{g}}\phi. \end{aligned}$$

Consequently, the sign of $\lambda_1(g)$ is a conformal invariant. In particular, by using an eigenfunction associated with $\lambda_1(L)$ as a conformal factor, it follows that $[g]$ admits a metric $\hat{g}$ with $\Phi_{\hat{g}} > 0$ (resp., $= 0$, < 0) if and only if $\lambda_1(L_g) > 0$ (resp., $= 0$, < 0).

One departure from the classical Yamabe problem is that the modified scalar curvature may only be Lipschitz continuous. Therefore, the Schauder estimates imply that the first

eigenfunction is in $C^{2,\alpha}$ and hence defines a conformal metric which is only $C^{2,\alpha}$. One can smooth $|F^+|$ and approximate, but in our setting this will not be necessary.

Now suppose that $F^+ \not\equiv 0$. Then from the inequality (9), $u^{1/2}$ satisfies

$$0 \geq L_g(|F^+|^{1/2}).$$

Multiplying by $|F^+|^{1/2}$ and integrating over M^4 gives

$$0 \geq \int_X |F^+|^{1/2} L_g(|F^+|^{1/2}) dV_g.$$

so the lowest eigenvalue $\lambda_1(L) \leq 0$.

Let $\phi > 0$ be the eigenfunction associated with λ_1 , and define $g' := \phi^2 g$. Then:

$$\Phi_{g'} = \phi^{-3} L\phi = \lambda_1 \phi^{-2} \leq 0.$$

Hence, in this conformal metric g' , the modified scalar curvature is nonpositive:

$$R_{g'} - 2\sqrt{6}|W^+|_{g'} - 3\gamma_1|F^+|_{g'} \leq 0.$$

Integrating over X yields

$$\int_X R_{g'} dV_{g'} \leq 2\sqrt{6} \int_X |W^+|_{g'} dV_{g'} + 3\gamma_1 \int_X |F^+|_{g'} dV_{g'}.$$

Using the Cauchy–Schwarz inequality and conformal invariance of the L^2 norms:

$$\int_X R_{g'} dV_{g'} \leq \left(2\sqrt{6}\|W^+\|_{L^2} + 3\gamma_1\|F^+\|_{L^2}\right) \cdot \text{Vol}(g')^{1/2}.$$

On the other hand, by the definition of the Yamabe invariant,

$$\int_X R_{g'} dV_{g'} \geq Y([g]) \cdot \text{Vol}(g')^{1/2}.$$

Combining these facts, we obtain

$$Y([g]) \leq 3\gamma_1\|F^+\|_{L^2} + 2\sqrt{6}\|W^+\|_{L^2},$$

which is precisely the inequality (1) claimed in the main theorem.

2.4 Equality Case and Geometric Consequences

Suppose equality holds in the main inequality:

$$Y([g]) = 3\gamma_1\|F^+\|_{L^2} + 2\sqrt{6}\|W^+\|_{L^2}.$$

We now explain why this leads to strong geometric consequences.

Step 1: Consequences of Cauchy–Schwarz equality

Equality in the Cauchy–Schwarz inequality implies that the functions involved must be proportional to constants:

$$|F^+| = \text{const}, \quad |W^+| = \text{const},$$

in the Yamabe-minimizing conformal metric g' . That is, the pointwise norms of the self-dual curvature and the Weyl tensor are constant over X^4 with respect to g' .

Step 2: Vanishing of ∇F^+

Returning to the Bochner formula in the Yamabe metric g' , and using that all scalar functions are constant, the inequality becomes:

$$\frac{1}{2}\Delta|F^+|^2 = |\nabla F^+|^2 \geq 0.$$

But $|F^+|$ is constant, so $\Delta|F^+|^2 = 0$. Hence,

$$|\nabla F^+|^2 = 0 \quad \Rightarrow \quad \nabla F^+ = 0.$$

Thus, F^+ is parallel with respect to the Levi-Civita connection of g' .

Step 3: Modified scalar curvature is zero

Recall the modified scalar curvature:

$$\Phi_{\hat{g}} = R_{\hat{g}} - 2\sqrt{6}|W^+|_{\hat{g}} - 3\gamma_1|F^+|_{\hat{g}}.$$

This quantity appeared in the definition of the modified Yamabe operator $L = -6\Delta + \Phi$. Equality in the energy inequality implies $\lambda_1(L) = 0$, and that L is minimized by a smooth, positive function (the Yamabe minimizer), so the scalar curvature in the conformal metric g' satisfies:

$$\Phi_{g'} \equiv 0.$$

Hence,

$$R_{g'} = 2\sqrt{6}|W^+|_{g'} + 3\gamma_1|F^+|_{g'}.$$

This is a pointwise identity, not just an integral one, and reflects a perfect balance between geometric and gauge curvature in the equality case.

Step 4: Topological implications from the Bochner formula

To understand the final part of the theorem—that either $b_2^+(X^4) = 0$ when $\gamma_1 > 0$, or that all harmonic self-dual 2-forms are parallel when $\gamma_1 = 0$ —we analyze real-valued self-dual 2-forms under the Yamabe metric g' .

Let $\omega \in \Lambda_+^2(X, \mathbb{R})$ be a harmonic self-dual 2-form. Then the classical Bochner-Weitzenböck Formula (3) gives

$$\frac{1}{2}\Delta|\omega|^2 = |\nabla\omega|^2 - 2\langle W^+ \star \omega, \omega \rangle + \frac{1}{3}R|\omega|^2.$$

From the estimate for the Weyl tensor term (5), we obtain

$$\frac{1}{2}\Delta|\omega|^2 \geq |\nabla\omega|^2 + \frac{1}{3}(R - 2\sqrt{6}|W^+|)|\omega|^2.$$

But in the equality case, the modified scalar curvature vanishes pointwise, i.e.

$$R - 2\sqrt{6}|W^+| = 3\gamma_1|F^+| > 0 \quad \text{if } \gamma_1 > 0.$$

Hence

$$\frac{1}{2}\Delta|\omega|^2 \geq |\nabla\omega|^2 + \gamma_1|F^+||\omega|^2. \quad (12)$$

Now integrate over X . Since X is closed, the integral of $\Delta|\omega|^2$ vanishes, then (12) reads

$$0 \geq \int_X |\nabla\omega|^2 + \gamma_1|F^+||\omega|^2.$$

If $\gamma_1 > 0$ and $|F^+| > 0$, this implies that both $|\nabla\omega|$ and $|\omega|$ must vanish, so $\omega \equiv 0$, i.e., there are no nontrivial harmonic self-dual 2-forms. Hence $b_2^+(X) = 0$.

Now consider the special case when $\gamma_1 = 0$. Then the inequality (12) becomes

$$\frac{1}{2}\Delta|\omega|^2 \geq |\nabla\omega|^2,$$

which implies that $|\nabla\omega| \equiv 0$, so ω is parallel.

3 Illustrative Example: The BPST Instanton on S^4

It is possible to achieve equality in (1) via the classic construction of BPST/ADHM SU(2) instantons on $\mathbb{S}^4$. The standard connection is expressed on $\mathbb{R}^4$, thought of as $\mathbb{S}^4 \setminus \{N\}$ as the $\mathfrak{su}(2)$ -valued 1-form

$$\theta = \frac{1}{1+|x|^2}(\theta_1 \otimes \mathbf{i} + \theta_2 \otimes \mathbf{j} + \theta_3 \otimes \mathbf{k}),$$

where

$$\begin{aligned} \theta_1 &= x_1 dx^2 - x_2 dx^1 + x_3 dx^4 - x_4 dx^3 \\ \theta_2 &= x_1 dx^3 - x_3 dx^1 + x_4 dx^2 - x_2 dx^4 \\ \theta_3 &= x_1 dx^4 - x_4 dx^1 + x_2 dx^3 - x_3 dx^2. \end{aligned}$$

and

$$\mathbf{i} = \left(\begin{array}{cc|c} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right), \quad \mathbf{j} = \left(\begin{array}{cc|c} 0 & 0 & 1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{array} \right), \quad \mathbf{k} = \left(\begin{array}{cc|c} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{array} \right).$$

A classic calculation yields the relevant curvature tensor

$$F_{\nabla} = \frac{2}{(1+|x|^2)^2} \{ (dx^{12} + dx^{34}) \otimes \mathbf{i} + (dx^{13} - dx^{24}) \otimes \mathbf{j} + (dx^{14} + dx^{23}) \otimes \mathbf{k} \}.$$

Using our conventions for the metric induced by the Euclidean metric on two-forms, $|dx^{12}|^2 = 2$, etc. Moreover, using the inner product on $\mathfrak{g}_E$, it follows that $|\mathbf{i}|^2 = |\mathbf{j}|^2 = |\mathbf{k}|^2 = 2$. Putting these together we can compute the pointwise norm of F_{∇} in this context to yield

$$|F_{\nabla}|^2 = \frac{4}{(1+|x|^2)^4} \left(2|dx^{12} + dx^{34}|^2 + 2|dx^{13} - dx^{24}|^2 + 2|dx^{14} + dx^{23}|^2 \right) = \frac{96}{(1+|x|^2)^4}.$$

Using the conformal invariance of the Yang–Mills energy, we thus obtain

$$\|F_{\nabla}\|_{L^2(\mathbb{S}^4, g_{\mathbb{S}^4})}^2 = 96 \int_{\mathbb{R}^4} \frac{1}{(1+|x|^2)^4} dV_{Eucl} = 16\pi^2.$$

Having computed the Yang–Mills energy, we turn to the remaining quantities in (1). A well-know fact is that the Yamabe invariant of the round 4-sphere is $8\sqrt{6}\pi$. Thus, employing Lie-Algebraic Estimate (6) we observe that the right hand side of the right hand side of (1) can be estimated as

$$3\gamma_1 \|F_{\nabla}^+\|_{L^2} \leq 3 \left(\frac{2}{\sqrt{3}} \gamma_0 \right) (4\pi) \leq 8\sqrt{6}\pi = Y([g_{\mathbb{S}^4}]).$$

Thus (1.1) is an equality, as are all the intermediate estimates.

The above calculation verifies that the BPST instanton on (S^4, g_{S^4}) saturates the energy gap inequality stated in Theorem 2.1. As a result, all intermediate estimates used in the proof must also be equalities. This rigid equality chain reflects several remarkable geometric features of the BPST/ADHM construction:

- First, the curvature of this $SU(2)$ instanton achieves equality in the key algebraic inequality involving the Lie bracket:

$$\langle \omega, [\omega, \omega] \rangle \leq \gamma_1 |\omega|^3,$$

realizing the optimal value $\gamma_1 = \frac{4\sqrt{3}}{3}$. This demonstrates the sharpness of the algebraic constant in the theorem.

- Second, the equality case forces *parallelism* of the self-dual curvature F_{∇}^+ with respect to a specific metric in the conformal class $[g]$. In this example, the distinguished BPST instanton connection has curvature that is already parallel with respect to the round metric g_{S^4} .

- Third, since all BPST/ADHM connections are related by conformal transformations (i.e., they lie in the same orbit under the conformal group action), it follows that any such instanton also has parallel curvature with respect to a metric in its conformal class. What the theorem reveals is that this metric arises naturally as the solution to a modified Yamabe problem—a variational method that not only proves the inequality but also constructs the distinguished conformal representative in a general setting.

In conclusion, this explicit example not only confirms the sharpness of the energy gap but also illustrates how geometric rigidity—parallelism, constant curvature norm, vanishing W^+ —emerges in the equality case. The BPST instanton serves as a canonical model demonstrating the deep link between Yang–Mills theory, conformal geometry, and scalar curvature optimization.

4 Applications and Corollaries

One immediate corollary is a universal energy bound for non-instanton Yang–Mills connections on conformally flat four-manifolds, that is

Corollary 4.1. *Let (X^4, g) be a closed, oriented, conformally flat four-manifold with $Y([g]) > 0$. Suppose ∇ is a Yang–Mills connection on a vector bundle E over X^4 with structure group $G \subset SO(E)$, and curvature F_∇ . Then ∇ is either an instanton, or satisfies*

$$\int_X |F_\nabla|_g^2 dV_g \geq 4\pi^2 p_1(E) + \frac{2Y([g])^2}{9\gamma_1^2} \geq 4\pi^2 p_1(E) + \frac{Y([g])^2}{12}. \quad (13)$$

In particular,

1. *For $E \rightarrow (\mathbb{S}^4, g_{\mathbb{S}^4})$ an $SU(2)$ bundle, a Yang–Mills connection ∇ is either an instanton, or satisfies*

$$\int_{\mathbb{S}^4} |F_\nabla|_g^2 dV_g \geq 4\pi^2 |p_1(E)| + 32\pi^2. \quad (14)$$

2. *For $E \rightarrow (\mathbb{S}^4, g_{\mathbb{S}^4})$ an $SO(3)$ bundle, a Yang–Mills connection ∇ is either an instanton, or satisfies*

$$\int_{\mathbb{S}^4} |F_\nabla|_g^2 dV_g \geq 4\pi^2 |p_1(E)| + 64\pi^2. \quad (15)$$

Proof. We recall the fundamental Chern–Weil formula

$$4\pi^2 p_1(E) = \int_X (|F_\nabla^-|_g^2 - |F_\nabla^+|_g^2) dV_g. \quad (16)$$

Since we have assumed our metric is conformally flat and $Y([g]) > 0$, we may combine (1) and (16) to yield

$$\begin{aligned} \|F_{\nabla}\|_{L^2}^2 &= \|F_{\nabla}^-\|_{L^2}^2 + \|F_{\nabla}^+\|_{L^2}^2 \\ &= 4\pi^2 p_1(E) + 2 \int_X |F_{\nabla}^+|_g^2 dV_g \\ &\geq 4\pi^2 p_1(E) + \frac{2Y([g])^2}{9\gamma_1^2}, \end{aligned}$$

as claimed. For the special case of $(\mathbb{S}^4, g_{\mathbb{S}^4})$, as discussed before we know that the Yamabe invariant is $8\sqrt{6}\pi$. Moreover using Lie-Algebraic Estimate part in section 2.2, we know that in the case of structure group $\mathrm{SU}(2)$ we may choose $\gamma_1 = \frac{4}{\sqrt{6}}$, whereas for $\mathrm{SO}(3)$ one has $\gamma_1 = \frac{2}{\sqrt{3}}$. This yields the remaining statements. $\square$

Another powerful application concerns the long-time behavior of the Yang–Mills flow. Building on prior work on bubbling and energy concentration, the authors show that if a connection on the trivial bundle has energy below $16\pi^2$, the flow must exist globally and converge to a flat connection. Their result gives an explicit, sharp energy threshold for global existence—something not previously achievable without stronger geometric assumptions.

5 Broader Significance and Future Directions

This work represents a significant conceptual advance in Yang–Mills theory. It not only bridges curvature estimates with conformal geometry but also introduces a new paradigm for understanding non-self-dual Yang–Mills connections through global invariants like the Yamabe constant. The conformally invariant nature of the result allows it to transcend the limitations of prior local curvature positivity assumptions.

In terms of future work, natural directions include extending this framework to complete noncompact manifolds, exploring whether analogous results exist in dimensions beyond four, and further leveraging the modified Yamabe operator in related geometric variational problems. Moreover, the connection to Yang–Mills flow dynamics opens pathways for deeper study of singularity formation and stability criteria in gauge theory.